Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
3	(a)			**Any 2 (×1) mark** Heart (1) Blood vessels/named blood vessels (1) Trachea/ windpipe (1) Bronchi/ bronchus (not bronchioles) (1) Oesophagus/ gullet (1) NOT ribs/ intercostal muscles/ diaphragm/ alveoli	2			2		
	(b)			Any 5 (×1) from • Diaphragm {contracts / flattens/ lowers/ moves down} (1) NOT constrict/ tightens/ straightens/ gets thinner • {Intercostal/ rib} muscles <u>contract</u> (1) • causing {ribs/ rib cage/chest wall} to move {out /up}/ chest expands (1) • {Volume/ space} of (thoracic cavity/ chest/ thorax) increases (1) • Pressure in (thoracic cavity/ chest/ thorax) decreases (1) • Lungs {inflate/ expand/ increase in size/ get bigger}/ air enters the lungs/ volume of lungs increases (1)	5			5		
	(c)	(i)		0.09 = 2 marks **If incorrect award 1 mark for:** $\frac{4.8}{5\,500} \times 100$ (method) 0.04 (not doubling volume) 0.0872727 (not 2 dp) 0.08 (rounding down)		2		2	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(ii)		Candidate must make very clear whether they are referring to the thorax ('in life') or the bell jar model. Any 2 for 1 mark each • balloons only occupy {0.04%/ 0.09%} of the space in the bell jar / balloons do not fill the jar/ ORA • Wall of bell jar is {rigid/ does not move}/ does not show the {<u>movement</u>/ <u>contraction</u>} of the intercostal muscles • At {rest/ expiration} rubber sheet is flat / diaphragm is domed}/ In inspiration rubber sheet pulls downwards further / diaphragm goes flat			2	2		
				Question 3 total	**7**	**2**	**2**	**11**	**2**	**0**